

ParaPro Assessment Mathematics Section — Exam 1

PART A — QUESTIONS

PARAPRO1-031

Section: Full

Type: mcq

Category: Mathematics

Skill: Whole-number multiplication

Prompt:

A teacher has 7 boxes of pencils. Each box contains 12 pencils. How many pencils are there in all?

A) 72

B) 84

C) 91

D) 96

PARAPRO1-032

Section: Full

Type: mcq

Category: Mathematics

Skill: Place value

Prompt:

In the number 38,472, what is the value of the digit 4?

A) 4

B) 40

C) 400

D) 4,000

PARAPRO1-033

Section: Full

Type: mcq

Category: Mathematics

Skill: Adding fractions

Prompt:

What is $\frac{3}{8} + \frac{1}{4}$?

A) $\frac{4}{12}$

B) $\frac{5}{8}$

C) $\frac{4}{8}$

D) $\frac{1}{2}$

PARAPRO1-034

Section: Full

Type: mcq

Category: Mathematics

Skill: Comparing decimals

Prompt:

Which decimal is greatest?

A) 0.7

B) 0.65

C) 0.708

D) 0.69

PARAPRO1-035

Section: Full

Type: mcq

Category: Mathematics

Skill: Finding a percent of a number

Prompt:

What is 20% of 45?

- A) 6
- B) 9
- C) 18
- D) 25

PARAPRO1-036

Section: Full

Type: mcq

Category: Mathematics

Skill: Order of operations

Prompt:

What is the value of $18 - 6 \div 3 \times 2$?

- A) 8
- B) 10
- C) 14
- D) 16

PARAPRO1-037

Section: Full

Type: mcq

Category: Mathematics

Skill: Solving simple equations

Prompt:

Solve for x: $x + 15 = 42$

- A) 27
- B) 37
- C) 57
- D) 630

PARAPRO1-038

Section: Full

Type: mcq

Category: Mathematics

Skill: Ratios and proportions

Prompt:

If 3 notebooks cost \$6, what is the cost of 5 notebooks at the same price per notebook?

- A) \$8
- B) \$9
- C) \$11
- D) \$10

PARAPRO1-039

Section: Full

Type: mcq

Category: Mathematics

Skill: Perimeter

Prompt:

A rectangular rug is 9 ft long and 4 ft wide. What is its perimeter?

A) 13 ft

B) 18 ft

C) 26 ft

D) 36 ft

PARAPRO1-040

Section: Full

Type: mcq

Category: Mathematics

Skill: Area

Prompt:

A bulletin board is 5 ft wide and 3 ft tall. What is its area?

A) 15 square ft

B) 16 square ft

C) 8 square ft

D) 30 square ft

PARAPRO1-041

Section: Full

Type: mcq

Category: Mathematics

Skill: Measurement conversion

Prompt:

How many inches are equal to 3 ft?

A) 24 in.

B) 36 in.

C) 30 in.

D) 48 in.

PARAPRO1-042

Section: Full

Type: mcq

Category: Mathematics

Skill: Elapsed time

Prompt:

A math lesson begins at 9:15 a.m. and ends at 10:05 a.m. How long is the lesson?

A) 40 min

B) 45 min

C) 50 min

D) 55 min

PARAPRO1-043

Section: Full

Type: mcq

Category: Mathematics

Skill: Money

Prompt:

A student buys a folder for \$2.35 and an eraser for \$1.20. If the student pays with a \$5 bill, how much change should the student receive?

- A) \$0.55
- B) \$1.35
- C) \$1.55
- D) \$1.45

PARAPRO1-044

Section: Full

Type: mcq

Category: Mathematics

Skill: Estimation

Prompt:

Which is the best estimate for $198 + 403$?

- A) 500
- B) 550
- C) 600
- D) 700

PARAPRO1-045

Section: Full

Type: mcq

Category: Mathematics

Skill: Number patterns

Prompt:

What is the next number in the pattern?

4, 8, 16, 32, ____

- A) 48
- B) 64
- C) 60
- D) 72

PARAPRO1-046

Section: Full

Type: mcq

Category: Mathematics

Skill: Division with remainders

Prompt:

A class has 29 students. The students are placed into groups of 4. How many students will be left over after making as many full groups of 4 as possible?

- A) 1
- B) 2
- C) 3
- D) 4

PARAPRO1-047

Section: Full

Type: mcq

Category: Mathematics

Skill: Subtracting decimals

Prompt:

What is $6.4 - 2.75$?

- A) 3.65
- B) 4.35
- C) 3.75
- D) 4.25

PARAPRO1-048

Section: Full

Type: mcq

Category: Mathematics

Skill: Equivalent fractions

Prompt:

Which fraction is equivalent to $\frac{2}{3}$?

- A) $\frac{3}{4}$
- B) $\frac{4}{9}$
- C) $\frac{6}{9}$
- D) $\frac{8}{10}$

PARAPRO1-049

Section: Full

Type: mcq

Category: Mathematics

Skill: Interpreting data

Prompt:

A class voted for a game to play during recess.

Game: Kickball, Tag, Jump Rope, Soccer

Votes: 8, 5, 3, 10

How many more students voted for soccer than for tag?

- A) 3
- B) 4
- C) 5
- D) 6

PARAPRO1-050

Section: Full

Type: mcq

Category: Mathematics

Skill: Multi-step arithmetic

Prompt:

A teacher has 36 stickers. She gives 4 stickers to each of 6 students. How many stickers does she have left?

- A) 10
- B) 12
- C) 14
- D) 18

PARAPRO1-051

Section: Full

Type: mcq

Category: Mathematics

Skill: Classroom application: place value support

Prompt:

A student writes 506 as “five hundred sixty.” Which explanation would best help the student understand the error?

- A) “The 0 means there are no tens, so 506 is five hundred six.”
- B) “The 0 should be ignored because it has no value.”
- C) “The 6 is in the tens place, so the number is five hundred sixty.”
- D) “The number should be rounded to 510.”

PARAPRO1-052

Section: Full

Type: mcq

Category: Mathematics

Skill: Classroom application: fraction error analysis

Prompt:

A student says that $\frac{1}{3} + \frac{1}{3} = \frac{2}{6}$. What is the best way to explain the mistake?

- A) “When denominators are the same, add only the numerators.”
- B) “Fractions can never be added if the denominators are the same.”
- C) “The answer should be $\frac{1}{6}$ because the numerators stay the same.”
- D) “The denominator should be added before the numerator.”

PARAPRO1-053

Section: Full

Type: mcq

Category: Mathematics

Skill: Classroom application: choosing a representation

Prompt:

A paraprofessional is helping students compare 0.4 and 0.35. Which representation would be most helpful?

- A) A hundreds chart or base-ten grid showing tenths and hundredths
- B) A clock showing hours and minutes
- C) A ruler marked only in whole inches
- D) A multiplication table for whole numbers

PARAPRO1-054

Section: Full

Type: mcq

Category: Mathematics

Skill: Classroom application: checking reasonableness

Prompt:

A student estimates that 49×6 is about 3,000. Which response best supports the student in checking reasonableness?

- A) “Round 49 to 50, then calculate 50×6 to get about 300.”
- B) “Round 49 to 100, then calculate 100×6 to get 600.”
- C) “Add $49 + 6$ to check the multiplication.”
- D) “Divide 49 by 6 because multiplication estimates are not useful.”

PARAPRO1-055

Section: Full

Type: mcq

Category: Mathematics

Skill: Classroom application: interpreting a student solution

Prompt:

A student solves $24 \div 6$ by drawing 6 equal groups and placing 4 dots in each group. Which statement best describes the student's work?

- A) The student found 24×6 .
- B) The student correctly showed that $24 \div 6 = 4$.
- C) The student should have drawn 24 groups of 6.
- D) The student showed $24 - 6 = 18$.

PARAPRO1-056

Section: Full

Type: mcq

Category: Mathematics

Skill: Classroom application: measurement support

Prompt:

Students are measuring strips of paper to the nearest inch. One strip is a little more than 7 in. long but less than 8 in. long. Which measurement is most reasonable to record to the nearest inch?

- A) 6 in.
- B) 7 in.
- C) 8 in.
- D) 15 in.

PARAPRO1-057

Section: Full

Type: mcq

Category: Mathematics

Skill: Classroom application: word-problem strategy

Prompt:

A student is solving this problem: "There are 5 tables with 6 students at each table. How many students are there in all?" Which equation matches the problem?

- A) $5 + 6 = 11$
- B) $6 - 5 = 1$
- C) $5 \times 6 = 30$
- D) $30 \div 5 = 6$

PARAPRO1-058

Section: Full

Type: mcq

Category: Mathematics

Skill: Classroom application: money reasoning

Prompt:

A student says that 3 quarters and 2 dimes are worth 32¢. Which statement best identifies the error?

- A) The student counted each quarter as 25¢ and each dime as 10¢.
- B) The student likely counted the number of coins instead of the total value.
- C) The student should subtract the dimes from the quarters.
- D) The student should count each dime as 25¢.

PARAPRO1-059

Section: Full

Type: mcq

Category: Mathematics

Skill: Classroom application: data interpretation

Prompt:

A paraprofessional is helping a small group read this table.

Color: Red, Blue, Green, Yellow

Number of students: 4, 7, 6, 3

Which question can be answered directly from the table?

- A) Which color was liked by the most students?
- B) Why did students choose blue?
- C) Which color will students choose next week?
- D) How many students were absent from class?

PARAPRO1-060

Section: Full

Type: mcq

Category: Mathematics

Skill: Classroom application: area versus perimeter

Prompt:

A student finds the area of a 6 ft by 2 ft rectangular mat by adding $6 + 2 + 6 + 2 = 16$. What should the student be reminded to do?

- A) Add all side lengths to find area.
- B) Multiply length by width to find area.
- C) Divide the longer side by the shorter side.
- D) Round both side lengths before calculating.

PART B — ANSWER KEY + EXPLANATIONS

PARAPRO1-031 — Correct: B — Correct Answer: 84 — Explanation: There are 7 boxes with 12 pencils in each box, so $7 \times 12 = 84$ pencils.

PARAPRO1-032 — Correct: C — Correct Answer: 400 — Explanation: In 38,472, the digit 4 is in the hundreds place, so its value is 400.

PARAPRO1-033 — Correct: B — Correct Answer: $\frac{5}{8}$ — Explanation: Rewrite $\frac{1}{4}$ as $\frac{2}{8}$. Then $\frac{3}{8} + \frac{2}{8} = \frac{5}{8}$.

PARAPRO1-034 — Correct: C — Correct Answer: 0.708 — Explanation: Comparing to thousandths gives 0.700, 0.650, 0.708, and 0.690. The greatest is 0.708.

PARAPRO1-035 — Correct: B — Correct Answer: 9 — Explanation: 20% means $\frac{1}{5}$. One-fifth of 45 is $45 \div 5 = 9$.

PARAPRO1-036 — Correct: C — Correct Answer: 14 — Explanation: Do division and multiplication first from left to right: $6 \div 3 \times 2 = 4$. Then $18 - 4 = 14$.

PARAPRO1-037 — Correct: A — Correct Answer: 27 — Explanation: Subtract 15 from both sides: $x = 42 - 15 = 27$.

PARAPRO1-038 — Correct: D — Correct Answer: \$10 — Explanation: If 3 notebooks cost \$6, each notebook costs \$2. Five notebooks cost $5 \times \$2 = \10 .

PARAPRO1-039 — Correct: C — Correct Answer: 26 ft — Explanation: Perimeter is the distance around the rectangle: $9 + 4 + 9 + 4 = 26$ ft.

PARAPRO1-040 — Correct: A — Correct Answer: 15 square ft — Explanation: Area of a rectangle is length $\times$ width. So $5 \times 3 = 15$ square ft.

PARAPRO1-041 — Correct: B — Correct Answer: 36 in. — Explanation: Each foot has 12 inches. Therefore, $3 \text{ ft} = 3 \times 12 = 36$ in.

PARAPRO1-042 — Correct: C — Correct Answer: 50 min — Explanation: From 9:15 to 10:00 is 45 minutes, and from 10:00 to 10:05 is 5 more minutes. The total is 50 minutes.

PARAPRO1-043 — Correct: D — Correct Answer: \$1.45 — Explanation: The total cost is $\$2.35 + \$1.20 = \$3.55$. The change is $\$5.00 - \$3.55 = \$1.45$.

PARAPRO1-044 — Correct: C — Correct Answer: 600 — Explanation: Round 198 to 200 and 403 to 400. Then $200 + 400 = 600$.

PARAPRO1-045 — Correct: B — Correct Answer: 64 — Explanation: Each number is multiplied by 2. So $32 \times 2 = 64$.

PARAPRO1-046 — Correct: A — Correct Answer: 1 — Explanation: $29 \div 4 = 7$ remainder 1 because $7 \times 4 = 28$ and $29 - 28 = 1$.

PARAPRO1-047 — Correct: A — Correct Answer: 3.65 — Explanation: Line up the decimals: $6.40 - 2.75 = 3.65$.

PARAPRO1-048 — Correct: C — Correct Answer: $\frac{6}{9}$ — Explanation: Multiplying the numerator and denominator of $\frac{2}{3}$ by 3 gives $\frac{6}{9}$.

PARAPRO1-049 — Correct: C — Correct Answer: 5 — Explanation: Soccer received 10 votes and tag received 5 votes. The difference is $10 - 5 = 5$.

PARAPRO1-050 — Correct: B — Correct Answer: 12 — Explanation: The teacher gives away $4 \times 6 = 24$ stickers. She has $36 - 24 = 12$ stickers left.

PARAPRO1-051 — Correct: A — Correct Answer: “The 0 means there are no tens, so 506 is five hundred six.” — Explanation: In 506, the 0 is in the tens place, so there are no tens. The number is read as five hundred six.

PARAPRO1-052 — Correct: A — Correct Answer: “When denominators are the same, add only the numerators.” — Explanation: For $\frac{1}{3} + \frac{1}{3}$, the denominator stays 3 and the numerators are added: $1 + 1 = 2$, so the answer is $\frac{2}{3}$.

PARAPRO1-053 — Correct: A — Correct Answer: A hundreds chart or base-ten grid showing tenths and hundredths — Explanation: A hundreds chart or base-ten grid can show 0.40 and 0.35 clearly, making it easier to compare the decimals.

PARAPRO1-054 — Correct: A — Correct Answer: “Round 49 to 50, then calculate 50×6 to get about 300.” — Explanation: Since 49 is close to 50, 49×6 should be close to $50 \times 6 = 300$, not 3,000.

PARAPRO1-055 — Correct: B — Correct Answer: The student correctly showed that $24 \div 6 = 4$. — Explanation: Dividing 24 into 6 equal groups gives 4 in each group, so $24 \div 6 = 4$.

PARAPRO1-056 — Correct: C — Correct Answer: 8 in. — Explanation: A length a little more than 7 inches but less than 8 inches is closer to 8 inches if it is past the halfway point between 7 and 8.

PARAPRO1-057 — Correct: C — Correct Answer: $5 \times 6 = 30$ — Explanation: There are 5 groups with 6 students in each group, so multiplication is used: $5 \times 6 = 30$.

PARAPRO1-058 — Correct: B — Correct Answer: The student likely counted the number of coins instead of the total value. — Explanation: Three quarters are 75¢ and two dimes are 20¢, for a total of 95¢. The answer 32¢ suggests the student did not correctly count the coin values.

PARAPRO1-059 — Correct: A — Correct Answer: Which color was liked by the most students? — Explanation: The table shows the number of students for each color, so it can be used to identify the greatest number.

PARAPRO1-060 — Correct: B — Correct Answer: Multiply length by width to find area. — Explanation: The student found the perimeter by adding all side lengths. To find area, multiply length by width: $6 \times 2 = 12$ square ft.